

Customer Behavior Analysis – Alfido Tech

1. Title Page

Customer Behavior Analysis – Alfido Tech Prepared by: Satya Shankar Yadav InternSpark
Data Analyst Internship

2. Introduction

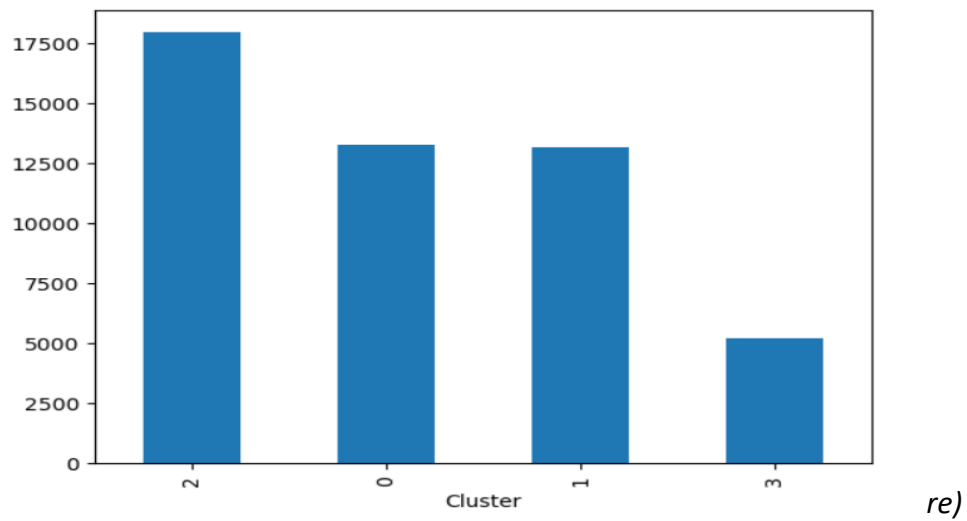
- **Dataset Source:** Kaggle – Customer Behavior Analysis
- **Goal:** Analyze customer transactions & behavior to identify segments, purchase patterns, and churn risks.
- **Deliverables:** Jupyter Notebook with code + PDF report summarizing findings and recommendations.

3. Data Cleaning & Feature Engineering

- Removed duplicates and handled missing values.
- Converted Purchase Date to datetime format.
- Created **RFM features**:
 - **Recency** – Days since last purchase
 - **Frequency** – Number of purchases per customer
 - **Monetary** – Total purchase amount

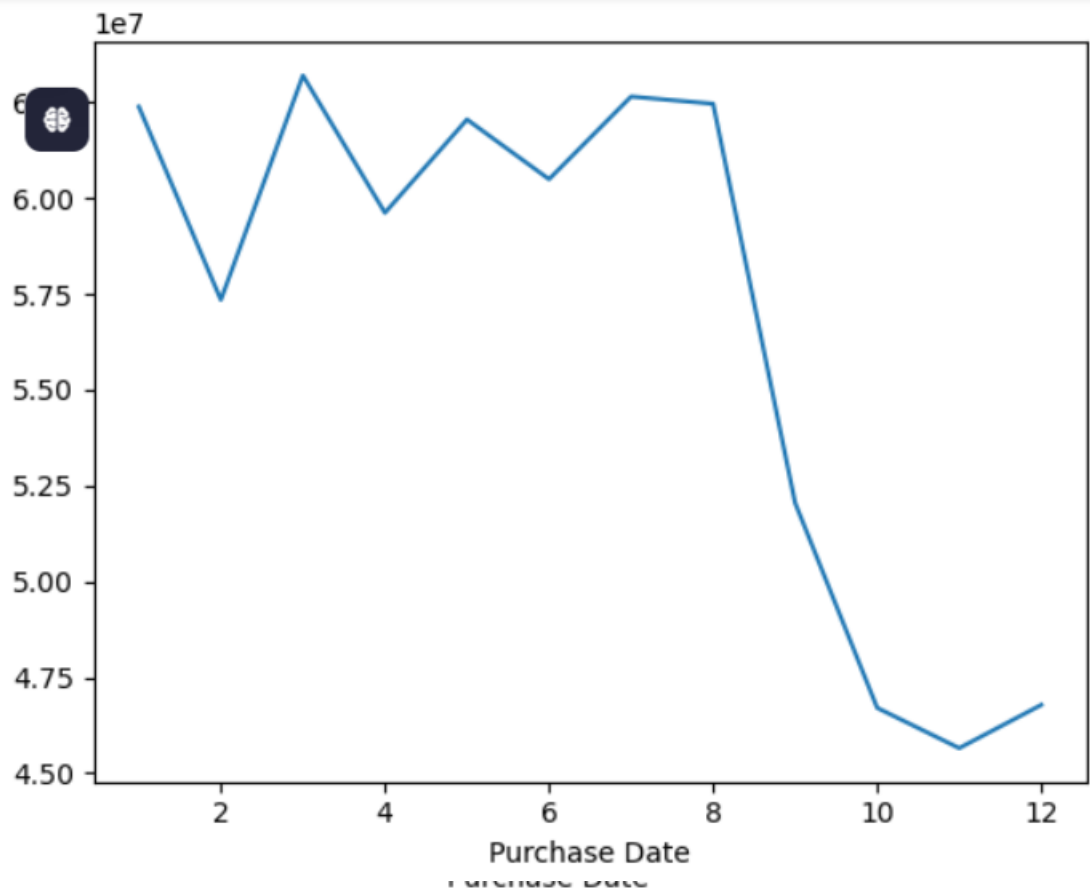
4. Customer Segmentation

- Applied **RFM scoring** and **KMeans clustering**.
- Segments identified:
 1. **High-Value Loyal Customers** – frequent buyers, high spend.
 2. **Young Frequent Buyers** – age 18–25, moderate spend, high frequency.
 3. **At-Risk Customers** – multiple returns, churn risk.
 4. **Low-Value Customers** – infrequent purchases, low spend.

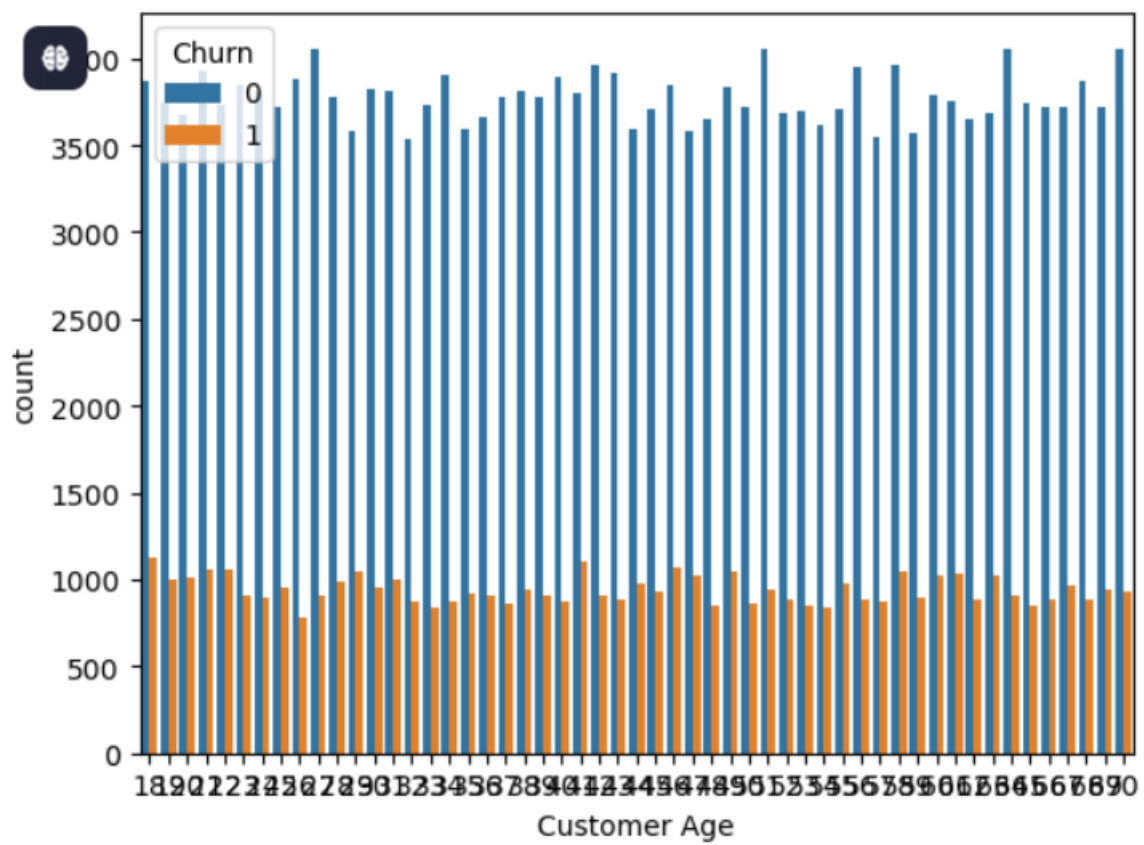


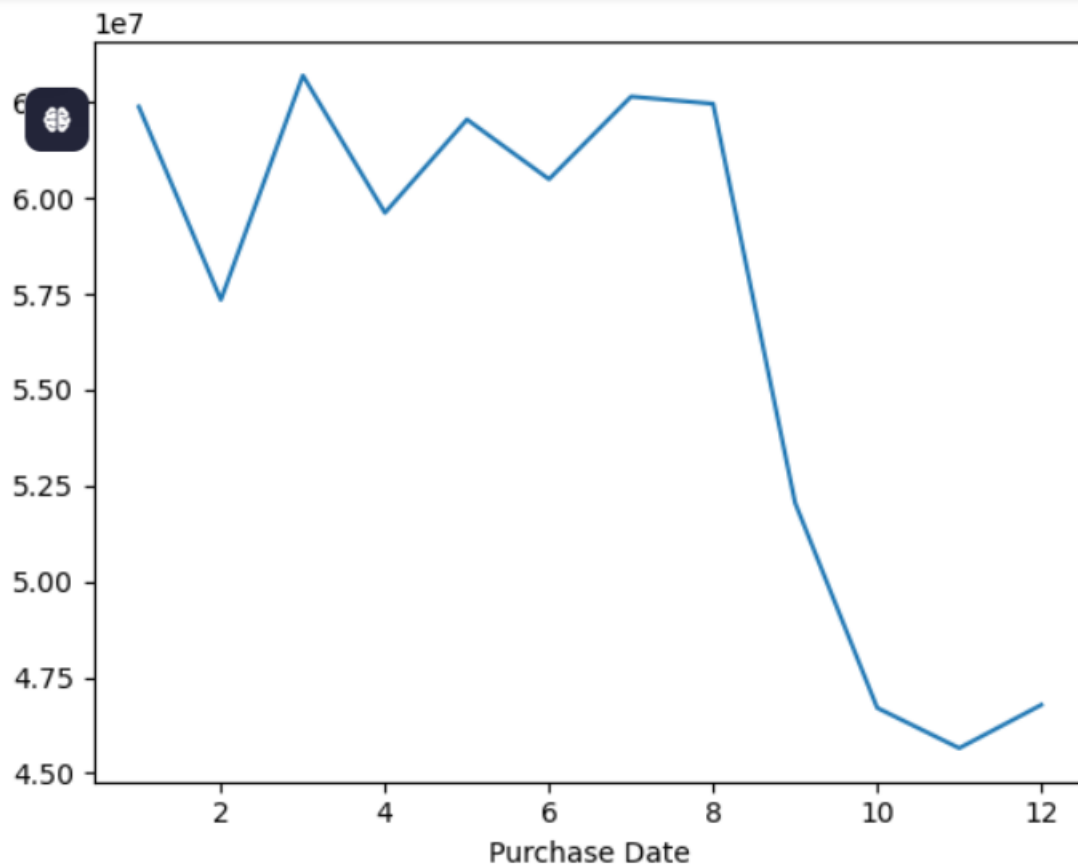
5. Purchase Patterns & Retention Trends

- **Category Trends:** Electronics & Fashion dominate sales.
- **Time Trends:** Weekend purchases peak consistently.
- **Churn Trends:** Customers with >2 returns show higher churn probability.



6.





Insights

1. 18–25 age group contributes highest purchase frequency.
2. Electronics & Fashion categories dominate total sales.
3. Weekend purchases peak consistently.
4. Customers with multiple returns show higher churn risk.
5. High-value loyal customers form ~20% of customer base.

7. Recommendations for Alfido Tech

1. Launch **loyalty rewards program** for high-value customers.
2. Run **weekend campaigns** to maximize peak sales.
3. Target **at-risk churn customers** with personalized offers.
4. Promote **cross-sell bundles** in Electronics/Fashion.
5. Use **segment-based ads** (social media for young, email for older).

8. Conclusion

This analysis highlights clear customer segments and purchase patterns. By implementing targeted engagement strategies, Alfido Tech can improve retention, reduce churn, and maximize revenue growth.